CST-250 Project Milestone Guide

Directions: Throughout this course, students will incrementally design and build a Minesweeper Game.

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Introduction

Coding Guidelines and Best Practices

The GCU College of Engineering and Technology Coding Guidelines and Best Practices for C#, available in the class resources, must be followed in all CST-250 projects to ensure consistency, readability, and professionalism in student code. These guidelines promote a clear separation of concerns by organizing code into the Models, BusinessLogicLayer, and DataAccessLayer folders. The document also outlines essential standards for casing, naming conventions, code structure, commenting, and organization. Additionally, it emphasizes best practices in object-oriented design, exception handling, and the use of design patterns. Adhering to these expectations is required for successful completion of course assignments.

What Is Minesweeper?

Minesweeper is a classic puzzle game that has been a common feature on personal computers for decades. The game combines logic and strategy: clear a grid of squares without hitting hidden mines.

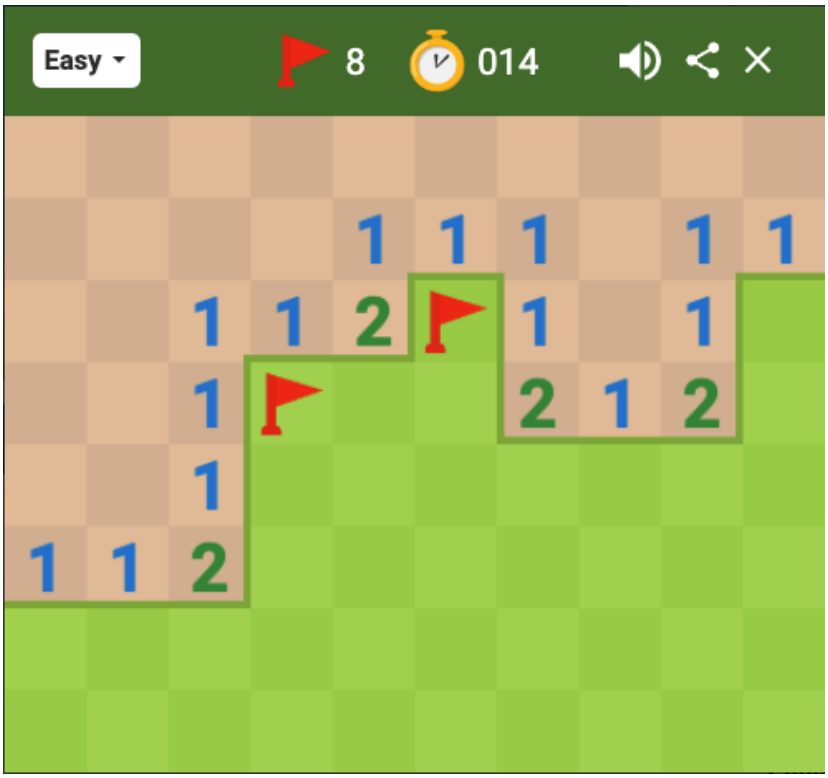


Figure 1. Online Minesweeper Game.

The gameplay involves a rectangular grid of tiles, each of which may conceal a mine. When a player clicks on a tile that does not contain a mine, the game reveals a number on that tile,

indicating how many adjacent tiles contain mines. This clue helps the player deduce which neighboring tiles are safe to click next.

Tiles adjacent to a mine can be marked with a flag to help the player keep track of suspected mines and prevent accidental clicks. The game continues until either all tiles without mines are revealed (resulting in a win) or a mine is triggered (resulting in a loss).

Minesweeper is not only a game of chance but a test of deductive reasoning and quick thinking. It offers various levels of difficulty, typically altering the size of the grid and the number of mines, allowing players from beginners to seasoned strategists to find a challenging yet manageable level.

Play the game here until you can regularly win the game. It will be impossible to create a Minesweeper app unless you are familiar with its rules.

Milestone 1: Console-Based Minesweeper Game Architecture and Initial Implementation

Overview

In this milestone, students will create the following.

Class Library Project

The class library must be organized using the **N-Layer Architecture** and include the following.

- **Models**
  - CellModel
  - BoardModel
  - **Note:** The model classes must not contain any business logic. They should only hold data definitions (properties and constructors, if necessary).
- **BusinessLogicLayer (BLL)**
  - All logic related to the behavior of the board and cells must reside in the BLL. No logic should be present in the model classes.

Console Application Project

- The Program.cs file must be used as the application entry point.
- **Do not use top-level statements**.
- Use **Activity 1** as a reference for correctly configuring the N-Layer Architecture and folder structure.

Object-Oriented Programming (OOP) Requirements

This milestone must demonstrate the following pillars of object-oriented programming:

- **Abstraction**
- **Inheritance**
- **Polymorphism**

Unit Testing Requirements

As part of Milestone 1, your solution must include a separate xUnit Test Project to verify the functionality of the business logic layer (BLL). You are required to implement xUnit tests that thoroughly exercise the methods and behaviors in your BLL. This ensures your logic is reliable, meets project requirements, and can be validated through automated testing. The

inclusion of these unit tests will also help guide development and support future maintenance of your application. Properly structured and passing xUnit tests are required for successful completion of this milestone.

Coding Standards

All code must adhere to the **GCU College of Engineering and Technology Coding Guidelines and Best Practices for C#**, which can be found in Class Rresources. This includes conventions for naming, code structure, commenting, etc.

Additional Guidance

Below is an example of what your team might include in the **Models** folder. Remember, **models should contain no logic**—they are strictly for representing the data structure used in the application.

1. UML Class Diagrams

Display the class properties and methods. A game cell should have the following properties:

a. **Row** and **Column**. These should initially be set to -1.

b. **IsVisited** Boolean value. This should initially be set to false. During the game, the player will "visit" or "clear" a square, changing the isVisited state from false to true.

c. **IsBomb** Boolean value. This should initially be set to false. "isLive" set to true will indicate that the cell is a live bomb cell.

d. **IsFlagged** Boolean value. This should initially be set to false. "isFlagged" set to true will indicate that the player has placed a flag on the cell. It may or may not be accurate to say that a flagged cell contains a bomb.

e. **NumberOfBombNeighbors**. The number of neighbors that contain bombs. This should initially be set to 0 but will change after running a special method that counts the surrounding bombs.

f. **HasSpecialReward**. Depending on your preference, a special reward cell can give the player a bonus when it is revealed. Choose one of these examples to implement as a reward. Some of these rewards sound fun but could actually make the game too easy to play, so choose carefully on what you want to give away with the special reward.

    i. **Hint** – Allows the player to use a "Show Bomb" option, revealing one bomb chosen at random.

    ii. **Time Freeze** – Stops the play clock for a period of time.

iii. **Bomb Defuse Kit** – Allows the player a free space click. If a player uses this on a space, a bomb will not end the game if present.

iv. **Bomb Squad** – Gives the player a bomb-defuse suite for 10 seconds of immunity. Any bombs clicked in the next 10 seconds will not end the game.

v. **Undo** – Provides the player with a mulligan, or a do-over.

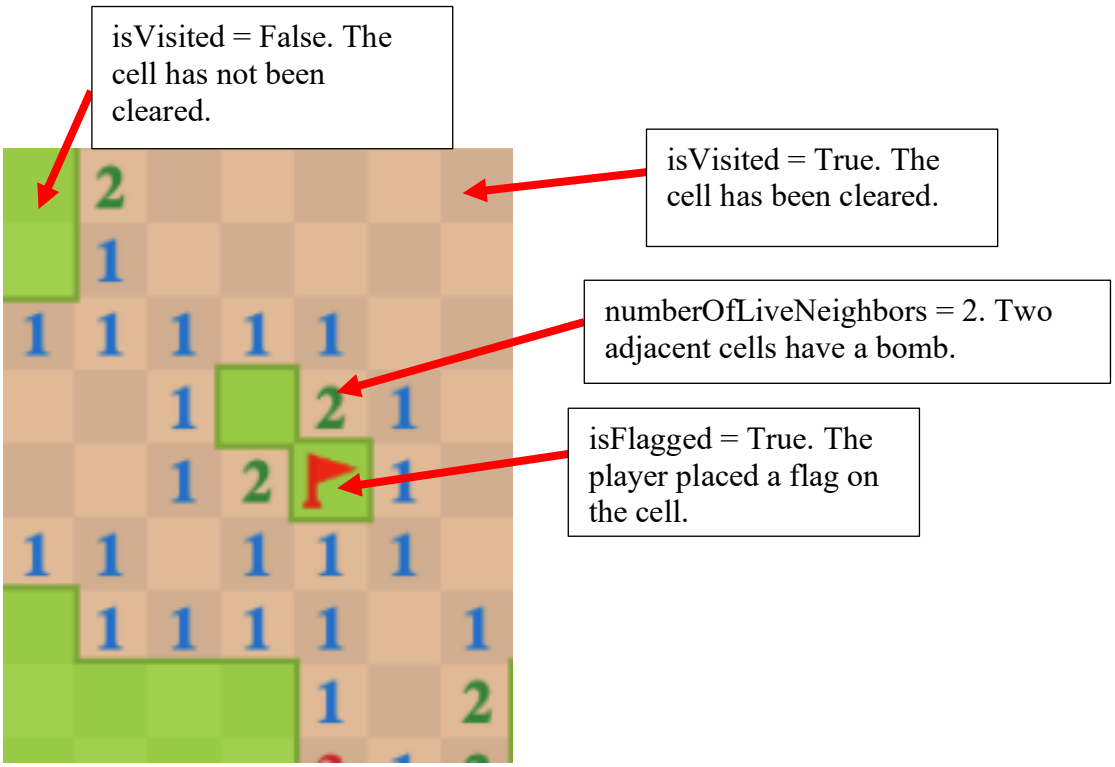


Figure 2. Cell properties will be displayed in the UI through various forms. The Cell class itself contains data in the form of integers and Boolean values.

g. Create a class that models a game board. A game board should have the following **properties**:

i. **Size**. The board will be square, where the size includes the dimensions of both the length and width of the board.

ii. **StartTime** – DateTime type. Stores the exact second that a game begins.

iii. **EndTime** – DateTime type. Stores the exact second that a game ends.

iv. **Cells** – The grid will be a 2-dimensional array of the type cell.

v. **Difficulty** – A number that represents how challenging it is to play the game. Decide from these options on how you want to determine the

number of bombs in the game. Any one of these options will work but will be programmed differently.

- **Difficulty Levels** – *Integer* (1, 2, 3) representing easy, medium, or hard. These levels will determine some number of bombs to be placed on the board.
- **Percentage** – *Float* (0.0 to 0.25) representing the probability that a cell will be assigned a bomb. Difficulty – 0.1 would indicate that 10% of cells will get bombs placed on them.
- **Count of Bombs** – *Integer* (10, 20, 30, etc.) representing the total number of bombs to be placed on the board.

vi. **RewardsRemaining** – Integer value of the number of special rewards the player has collected. Initially start with value of zero.

vii. **GameState** – Enum property to indicate the state of the game (e.g., **InProgress**, **Won**, **Lost**). This can be used to handle different outcomes and update the UI accordingly.

h. **All game board methods that involve logic must reside in the BusinessLogicLayer (BLL).** The BoardModel should only define the structure of the board, not how it functions.

i. The **constructor** for the Board should have a single parameter to set the size of the Grid. In its constructor, the Grid should be initialized so that a Cell object is stored at each location.

ii. **SetupBombs** – A method to randomly initialize the grid with live bombs. The method should utilize the difficulty property to determine which cells in the grid will be set to live status.

iii. **CountBombsNearby** – A method to calculate the live neighbors for every cell on the grid. A cell should have between 0 and 8 live neighbors. If a cell

itself is live, then you can set the neighbor count to 9.

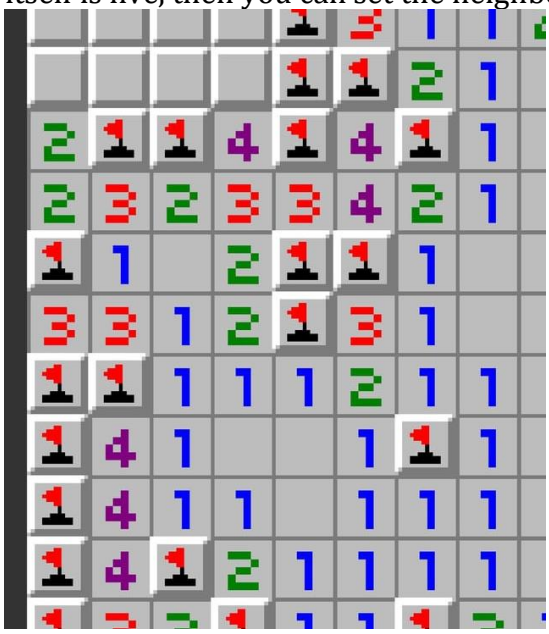


Figure 3. Game play reveals the "LiveNeighbor" values as color coded integers. In this case, most of the bombs have been successfully flagged by the player. Four is the maximum number of bombs adjacent to any square in this game. Many squares have zero bombs adjacent, which is represented by a blank gray square, not the number 0.

iv. **DetermineGameState** – Based on the state of the game, the method will return "GameLost," "GameWon," or "StillPlaying" status. We will not implement this method until later milestone assignments.

v. **UseSpecialBonus** – Depending on the special reward definition you choose in the Cell class, this method will put the reward into practice. We will not implement this method until later milestone assignments.

vi. **DetermineFinalScore** – This method will use a formula to calculate a score. The formula will be your invention, taking into account the difficulty level, the time elapsed, the size of the board, and special bonuses used. We will not implement this method until later milestone assignments.

2. UML Diagrams

a. UML diagrams must clearly illustrate a **composition relationship** between the appropriate classes.

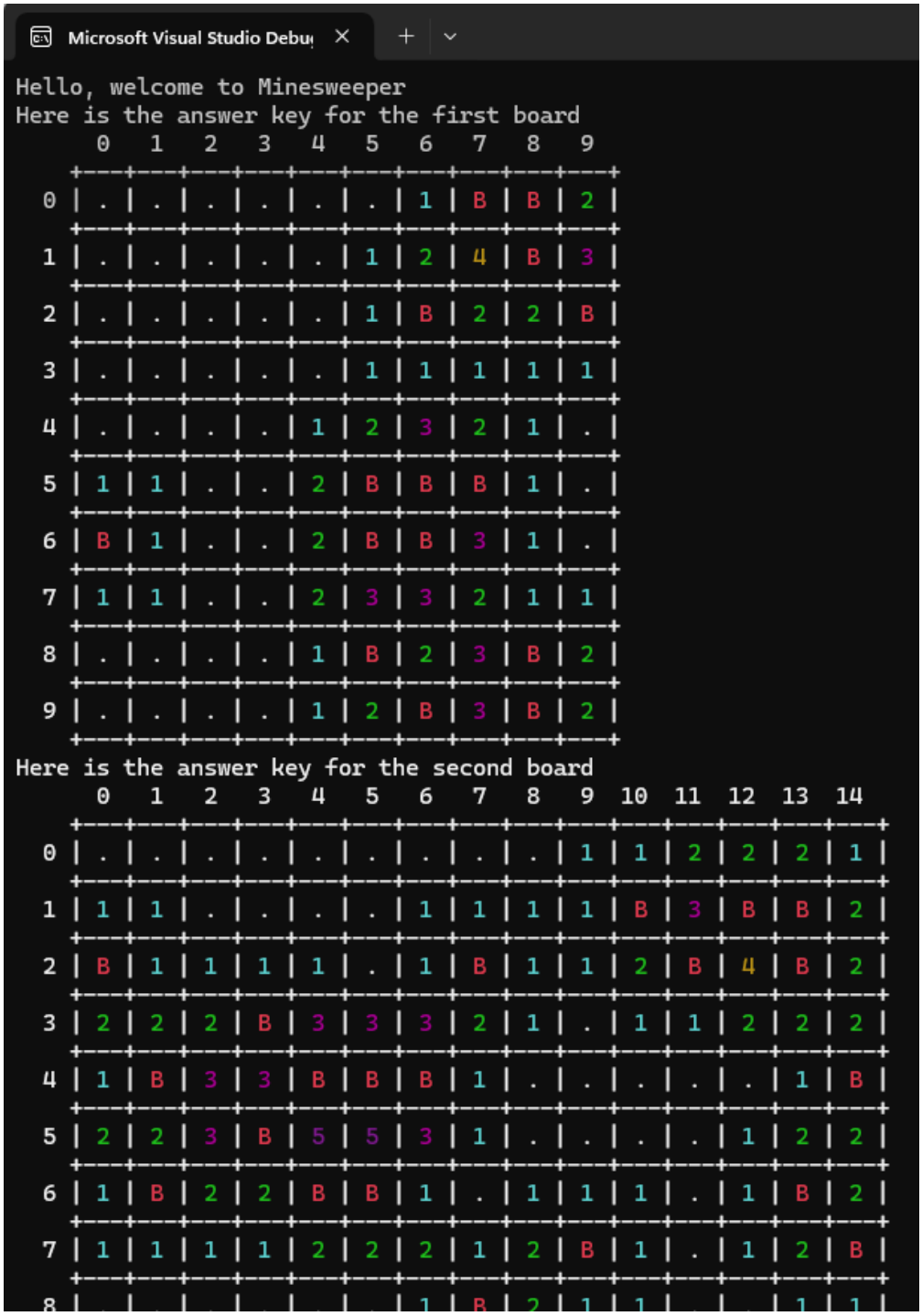
b. You may use any diagramming tool such as Visio**,** Draw.io, or similar software to create your UML diagram.

c. Once complete, insert the diagram into a Microsoft Word document and include it as part of your submission.

Additional Guidance

Below is an example of what your team might include in Program.cs. Remember, keep the entry point using clean code and move any code into functions or helper class methods that print to the console. When not using top-level statements there should not be a main method in Program.cs just like as shown in Activity 1.

a. Create a project reference to the class library project.
b. Create an instance of the BoardModel class in Program.cs.
c. The program should have a function or helper class method that **PrintAnswers(Board board)** to displays the contents of the Board.
d. Ensure that the board can print a variety of sizes of game boards. Test the output with a board size 10 or less and a second board greater than size 10.
e. Print "B" for bombs. Print a number if there are any neighboring bombs. Print a "." when there are no neighbor bombs.
f. Print column and row numbers. Print divider lines to display an outline around each cell.
g. Use Console.ForegroundColor to change the print color for each symbol. For example, this sets the print color to red. `Console.ForegroundColor = ConsoleColor.Red;`



```
Microsoft Visual Studio Debu
Hello, welcome to Minesweeper
Here is the answer key for the first board
      0   1   2   3   4   5   6   7   8   9
    +---+---+---+---+---+---+---+---+---+---+
  0 | . | . | . | . | . | . | 1 | B | B | 2 |
    +---+---+---+---+---+---+---+---+---+---+
  1 | . | . | . | . | . | 1 | 2 | 4 | B | 3 |
    +---+---+---+---+---+---+---+---+---+---+
  2 | . | . | . | . | . | 1 | B | 2 | 2 | B |
    +---+---+---+---+---+---+---+---+---+---+
  3 | . | . | . | . | . | 1 | 1 | 1 | 1 | 1 |
    +---+---+---+---+---+---+---+---+---+---+
  4 | . | . | . | . | 1 | 2 | 3 | 2 | 1 | . |
    +---+---+---+---+---+---+---+---+---+---+
  5 | 1 | 1 | . | . | 2 | B | B | B | 1 | . |
    +---+---+---+---+---+---+---+---+---+---+
  6 | B | 1 | . | . | 2 | B | B | 3 | 1 | . |
    +---+---+---+---+---+---+---+---+---+---+
  7 | 1 | 1 | . | . | 2 | 3 | 3 | 2 | 1 | 1 |
    +---+---+---+---+---+---+---+---+---+---+
  8 | . | . | . | . | 1 | B | 2 | 3 | B | 2 |
    +---+---+---+---+---+---+---+---+---+---+
  9 | . | . | . | . | 1 | 2 | B | 3 | B | 2 |
    +---+---+---+---+---+---+---+---+---+---+
Here is the answer key for the second board
      0   1   2   3   4   5   6   7   8   9  10  11  12  13  14
    +---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+
  0 | . | . | . | . | . | . | . | . | . | 1 | 1 | 2 | 2 | 2 | 1 |
    +---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+
  1 | 1 | 1 | . | . | . | . | 1 | 1 | 1 | 1 | B | 3 | B | B | 2 |
    +---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+
  2 | B | 1 | 1 | 1 | 1 | . | 1 | B | 1 | 1 | 2 | B | 4 | B | 2 |
    +---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+
  3 | 2 | 2 | 2 | B | 3 | 3 | 3 | 2 | 1 | . | 1 | 1 | 2 | 2 | 2 |
    +---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+
  4 | 1 | B | 3 | 3 | B | B | B | 1 | . | . | . | . | . | 1 | B |
    +---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+
  5 | 2 | 2 | 3 | B | 5 | 5 | 3 | 1 | . | . | . | . | 1 | 2 | 2 |
    +---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+
  6 | 1 | B | 2 | 2 | B | B | 1 | . | 1 | 1 | 1 | . | 1 | B | 2 |
    +---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+
  7 | 1 | 1 | 1 | 1 | 2 | 2 | 2 | 1 | 2 | B | 1 | . | 1 | 2 | B |
    +---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+
  8 | . | . | . | . | . | . | 1 | B | 2 | 1 | 1 | . | . | 1 | 1 |
```

Figure 4. Expected output for two game boards. The first is size 10. The second is size 15.

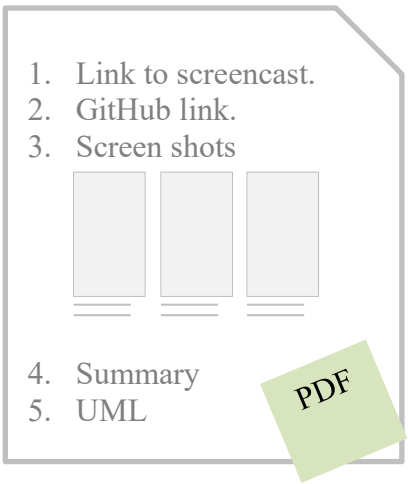
1. The main() method should do the following:
    a. Create an instance of the Board class.
    b. Call the Board.SetupBombs and Board. CountBombsNearby commands to initialize the grid.

c. Call the PrintAnswers(board) method to display the contents of the grid.

GitHub

1. Review the "GitHub Tutorial," located in Class Resources, to learn how to create a git repository and use a version control system.
2. Submit the current milestone to a GitHub.com private repository.
3. Provide access to your instructor to view the repository.
4. Include the URL for the repository in the project Microsoft Word document.

Deliverables



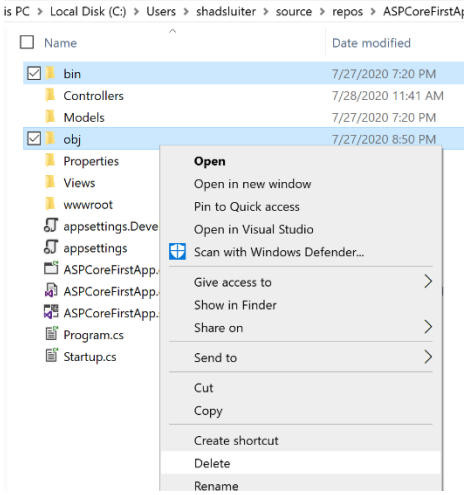
You must submit two separate files:

- One PDF containing Parts 1–7.
- One ZIP file containing the complete codebase for Part 8.

Milestone Submission

1. Create a cover page.
    a. Use APA formatting for the cover page.
2. Create a screencast video of the application.
    a. Demonstrate the application working per the requirements.
    b. Show relevant sections of the code, explaining the key ideas of this milestone and how they were implemented.
    c. This milestone must demonstrate the following pillars of object-oriented programming:
        i. Abstraction
        ii. Inheritance
        iii. Polymorphism
    d. Milestone must demonstrate the use of xUnit tests for BLL and must match UML diagrams.
    e. Paste a link to the video in a Microsoft Word document at the top of the second page.
    f. Do not just put a link to the video; there must be text that describes what the link is.
3. In the same Word document, add the link to the private GitHub repository below the screencast video link at the top of the second page. Share the repository with your instructor's email address.

4. In the same Word document, paste the UML diagrams for the classes and updated flowchart.
5. In the same Word document, add screenshots of the application being run. Show each screen of the output and put a caption under each picture explaining what is being demonstrated.
6. In the same document, in one paragraph, write a summary of the key concepts that were demonstrated in this lesson. Be sure to explain the key words introduced in this lesson.
7. Convert the Word document to a PDF and attach the document separately from the ZIP file. Multiple files can be uploaded with an assignment.
8. Submit a ZIP file of the project file. In order to save space, you can delete the bin and obj folders of the project. These folders contain the compiled version of the application and are automatically regenerated each time the build or run commands are executed.



Milestone 2: Interactive Console Version of Minesweeper with User Interaction

Overview

In this milestone, students will create an interactive playable version of the Minesweeper game.

1. Update the UML class diagram based on the new features described below.
2. In **Board.cs**, implement the method **DetermineGameState**. This method will inspect the contents of the Board and return one of three values:
    - **Won** when all non-bomb cells have been either visited or correctly flagged.
    - **Lost** if a bomb cell has been visited.
    - **StillPlaying** if there is at least one cell left to visit.
3. In **Program.cs**, create a method called **PrintBoard** that displays the board during the game play. The only difference between **PrintBoard** and **ShowAnswers** is that PrintBoard takes into account the **IsVisited** property of each cell. If a cell is not visited, display a "?" symbol.

Object-Oriented Programming (OOP) Requirements

This milestone must demonstrate the following pillars of object-oriented programming:

- Abstraction
- Encapsulation

Game Loop

In the main method of Program.cs, create a playable version of the game in a while.

Use this pseudocode as a guide.

```
ShowAnswers() // technically this is cheating to do this first.
bool victory = false; // player wins
bool death = false; // player loses
// repeat until the game is over
while (!victory && !death) {
        1. Ask the user for a row and column number.
        2. Ask if this is for "Flag" or "Visit" or "Use Reward."
        3. Update the status of the chosen cell according to the move.
        4. Determine if victory == true or if death == tru  Board.cs and Program.cs e
        5. Print the grid.
```

} // end game

Print a victory or failure message

```
C:\Users\shadsluiter\source\r

Hello, welcome to Minesweeper
Here is the answer key for the first board
      0   1   2   3
    +---+---+---+---+
  0 | 1 | 2 | 1 | 1 |
    +---+---+---+---+
  1 | B | 2 | B | 1 |
    +---+---+---+---+
  2 | 1 | 2 | 1 | 1 |
    +---+---+---+---+
  3 | . | . | . | . |
    +---+---+---+---+
Here is the current board
      0   1   2   3
    +---+---+---+---+
  0 | ? | ? | ? | ? |
    +---+---+---+---+
  1 | ? | ? | ? | ? |
    +---+---+---+---+
  2 | ? | ? | ? | ? |
    +---+---+---+---+
  3 | ? | ? | ? | ? |
    +---+---+---+---+
Enter the row number
0
Enter the column number
0
Enter 1 to visit the cell, 2 to flag the cell
1
Game in progress
```

```
Here is the current board
      0   1   2   3
    +---+---+---+---+
  0 | 1 | ? | ? | ? |
    +---+---+---+---+
  1 | ? | ? | ? | ? |
    +---+---+---+---+
  2 | ? | ? | ? | ? |
    +---+---+---+---+
  3 | ? | ? | ? | ? |
    +---+---+---+---+
Enter the row number
0
Enter the column number
1
Enter 1 to visit the cell, 2 to flag the cell
1
Game in progress
Here is the current board
      0   1   2   3
    +---+---+---+---+
  0 | 1 | 2 | ? | ? |
    +---+---+---+---+
  1 | ? | ? | ? | ? |
    +---+---+---+---+
  2 | ? | ? | ? | ? |
    +---+---+---+---+
  3 | ? | ? | ? | ? |
    +---+---+---+---+
Enter the row number
```

Figure 5. Game in progress. First, the answer key is shown. Then, two moves were attempted at 0,0 and 0,1.

```
Game in progress
Here is the current board
      0   1   2   3
    +---+---+---+---+
  0 | 1 | 2 | 1 | 1 |
    +---+---+---+---+
  1 | ? | 2 | ? | 1 |
    +---+---+---+---+
  2 | 1 | 2 | 1 | 1 |
    +---+---+---+---+
  3 | . | . | ? | ? |
    +---+---+---+---+
Enter the row number
3
Enter the column number
2
Enter 1 to visit the cell, 2 to flag the cell
1
Game in progress
Here is the current board
      0   1   2   3
    +---+---+---+---+
  0 | 1 | 2 | 1 | 1 |
    +---+---+---+---+
  1 | ? | 2 | ? | 1 |
    +---+---+---+---+
  2 | 1 | 2 | 1 | 1 |
    +---+---+---+---+
  3 | . | . | . | ? |
    +---+---+---+---+
Enter the row number
3
Enter the column number
3
Enter 1 to visit the cell, 2 to flag the cell
1
Congratulations! You won!
```

Figure 6. After placing 14 visit moves, the game is over and the player wins!

The Reward

Implement the code to make the reward bonus work properly.

In this example, the cell at 1,1 is a special reward cell. You can see the *r* character was chosen to display the reward.

```
Hello, welcome to Minesweeper
Here is the answer key for the first board
      0   1   2   3
    +---+---+---+---+
  0 | 1 | B | 2 | B |
    +---+---+---+---+
  1 | 2 | r | 3 | 1 |
    +---+---+---+---+
  2 | 1 | B | 2 | 1 |
    +---+---+---+---+
  3 | 1 | 1 | 2 | B |
    +---+---+---+---+
Here is the current board
```

Figure 7. Cell 1,1 was randomly chosen to contain the reward.

When the player visits cell 1,1, the reward message is shown.

```
Here is the current board
      0   1   2   3
    +---+---+---+---+
  0 | ? | ? | ? | ? |
    +---+---+---+---+
  1 | ? | ? | ? | ? |
    +---+---+---+---+
  2 | ? | ? | ? | ? |
    +---+---+---+---+
  3 | ? | ? | ? | ? |
    +---+---+---+---+
Enter the row number
1
Enter the column number
1
Enter 1 to visit the cell, 2 to flag the cell, 3 use a reward (bomb detector)
1
You found a reward!
```

Figure 8. Visiting 1,1 gets the reward.

On the next turn, the player uses the reward on cell 0,1. The reward allows the player peek to see if the cell is a bomb or not.

```
Enter the row number
0
Enter the column number
1
Enter 1 to visit the cell, 2 to flag the cell, 3 use a reward (bomb detector)
3
Is it a bomb? True
Game in progress
Here is the current board
      0   1   2   3
    +---+---+---+---+
  0 | ? | ? | ? | ? |
    +---+---+---+---+
  1 | ? | 2 | ? | ? |
    +---+---+---+---+
  2 | ? | ? | ? | ? |
    +---+---+---+---+
  3 | ? | ? | ? | ? |
    +---+---+---+---+
Enter the row number
```

Figure 9. Option 3 is chosen, which uses the reward. The player gets information about cell 0,1.

GitHub

1. Review the "GitHub Tutorial," located in Class Resources, to learn how to create a git repository and use a version control system.
2. Create a commit of this milestone and push it to the GitHub repository.
3. Your instructor will see the updates if you provided access.
4. Include the URL for the repository in the project Microsoft Word document.

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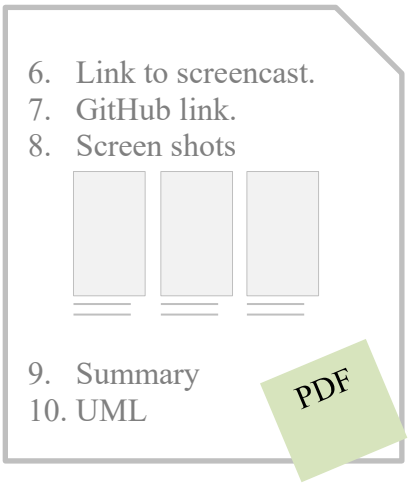

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    d. Milestone must demonstrate the use of xUnit tests for BLL and must match UML diagrams.
    e. Paste a link to the video in a Microsoft Word document at the top of the second page.
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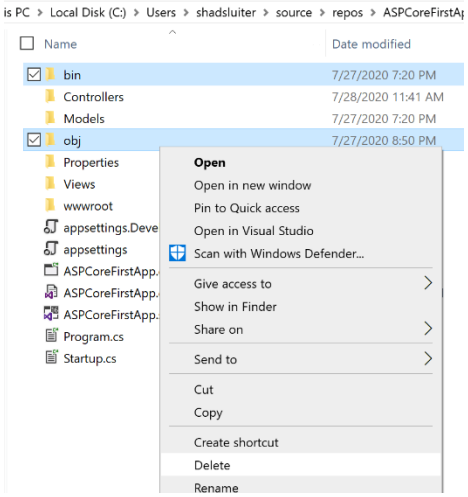
5. In the same Word document, add screenshots of the application being run. Show each screen of the output and put a caption under each picture explaining what is being demonstrated.



6. In the same document, in one paragraph, write a summary of the key concepts that were demonstrated in this lesson. Be sure to explain the key words introduced in this lesson.
7. Convert the Word document to a PDF and attach the document separately from the ZIP file. Multiple files can be uploaded with an assignment.
8. Submit a ZIP file of the project file. In order to save space, you can delete the bin and obj folders of the project. These folders contain the compiled version of the application and are automatically regenerated each time the build or run commands are executed.

Milestone 3: Recursive Flood Fill Implementation in Minesweeper

Overview

In this milestone, students will use recursion to develop an algorithm that reveals blocks of cells with no live neighbors.

1. Write FloodFill(int row, int col) method in Board for an algorithm that reveals blocks of cells with no live neighbors in a Minesweeper game. The images below demonstrate the behavior of this algorithm.

```
Hello, welcome to Minesweeper
Here is the answer key for the first board
      0   1   2   3   4   5
    +---+---+---+---+---+---+
  0 | . | . | . | . | 1 | B |
    +---+---+---+---+---+---+
  1 | . | . | . | . | 1 | 1 |
    +---+---+---+---+---+---+
  2 | . | 1 | 1 | 1 | . | . |
    +---+---+---+---+---+---+
  3 | . | 1 | B | 1 | . | . |
    +---+---+---+---+---+---+
  4 | r | 2 | 2 | 1 | . | . |
    +---+---+---+---+---+---+
  5 | 1 | B | 1 | . | . | . |
    +---+---+---+---+---+---+
```

Figure 10. Start of the game. Answers revealed for testing purposes.

```
Here is the current board
      0   1   2   3   4   5
    +---+---+---+---+---+---+
  0 | ? | ? | ? | ? | ? | ? |
    +---+---+---+---+---+---+
  1 | ? | ? | ? | ? | ? | ? |
    +---+---+---+---+---+---+
  2 | ? | ? | ? | ? | ? | ? |
    +---+---+---+---+---+---+
  3 | ? | ? | ? | ? | ? | ? |
    +---+---+---+---+---+---+
  4 | ? | ? | ? | ? | ? | ? |
    +---+---+---+---+---+---+
  5 | ? | ? | ? | ? | ? | ? |
    +---+---+---+---+---+---+
Enter the row number
0
Enter the column number
0
Enter 1 to visit the cell, 2 to flag the cell, 3 use a reward (bomb detector)
1
```

Figure 11. First move is 0,0.

```
Game in progress
Here is the current board
      0   1   2   3   4   5
    +---+---+---+---+---+---+
  0 | . | . | . | . | 1 | ? |
    +---+---+---+---+---+---+
  1 | . | . | . | . | 1 | 1 |
    +---+---+---+---+---+---+
  2 | . | 1 | 1 | 1 | . | . |
    +---+---+---+---+---+---+
  3 | . | 1 | ? | 1 | . | . |
    +---+---+---+---+---+---+
  4 | 1 | 2 | 2 | 1 | . | . |
    +---+---+---+---+---+---+
  5 | ? | ? | 1 | . | . | . |
    +---+---+---+---+---+---+
Enter the row number
```

Figure 12. Flood fill reveals most of the board. A few ? squares remain that were not touched by flood fill.

2. This flood fill method should be recursive. Inside the flood fill method, mark cells as "visited" = true when they are included in the block of affected cells. Recursively call floodfill on surrounding cells. You may wish to review the "FloodFill Animations Presentation," located in the Topic 3 activity.
3. Update the UML class diagram to include the new method.

GitHub

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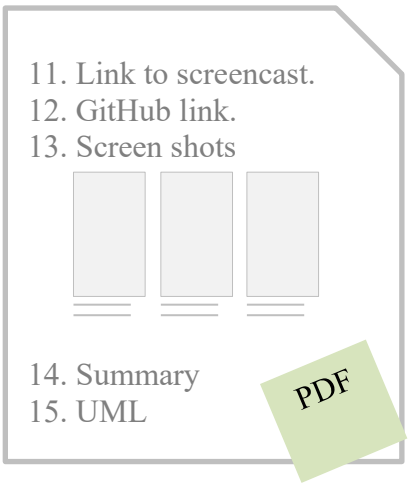
2. Create a commit of this milestone and push it to the GitHub repository.
3. Your instructor will see the updates if you provided access.
4. Include the URL for the repository in the project Microsoft Word document.

Deliverables



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Milestone Submission

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    a. Demonstrate the application working per the requirements.
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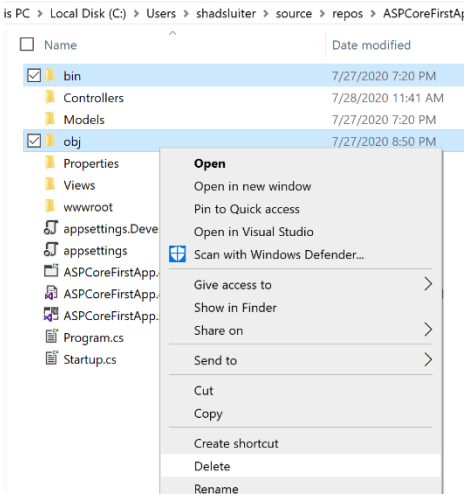

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Milestone 4: Minesweeper GUI Development with Enhanced User Interaction

Overview

In this milestone, students will create a GUI version of the Minesweeper game.

1. Add a new project to the solution. This project is a WinForms app.
2. Add a second form to get the game size and difficulty values from the user. Here are two suggested layouts for getting size and difficulty values from the user. Your form may need to be different depending on the properties of the Board class.

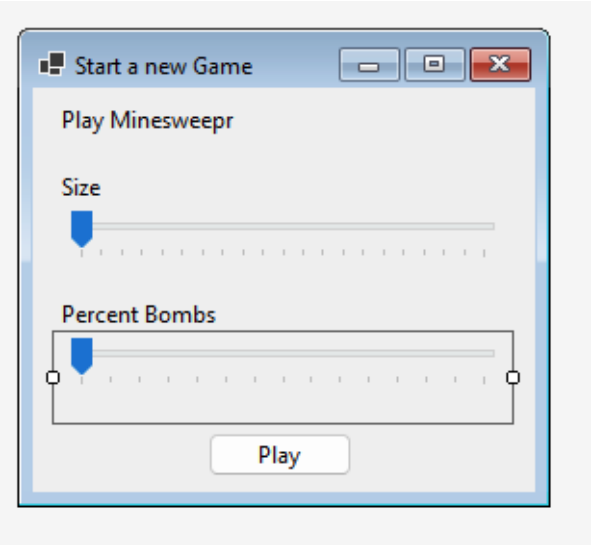

Figure 13. Two trackbars to get size and difficulty level.

Figure 14. Three radio buttons used to get starting difficulty from user.

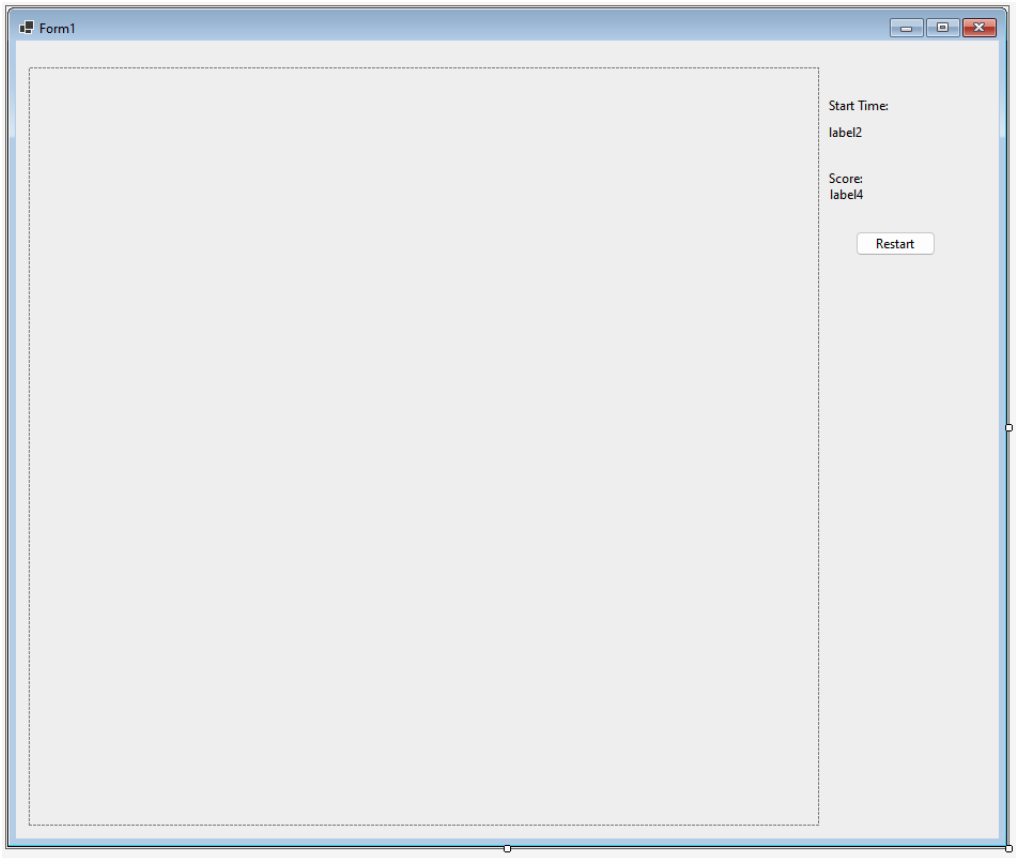

Figure 15. Suggested layout for the game form. A panel is used to contain the grid of buttons. Labels are used to display messages. A button is used to close the form and return to the start form.

3. Pass the data to the main game form to begin the game.
4. Using the Chessboard app from Activity 2, create a method to update the button faces based on the contents of the board.Cells properties. Use this suggested scheme or make one up.
    a. Print an "F" for a flagged cell.
    b. Print a "B" for a bomb.
    c. Print a blank or a number to indicate the number of neighboring bombs.
    d. Set the button background color to gray if it has not been visited yet.
5. Update the Program.cs file in the WinForms project. The startup form is shown on line 14. By default, the startup form is Form1.
6. Test the game.

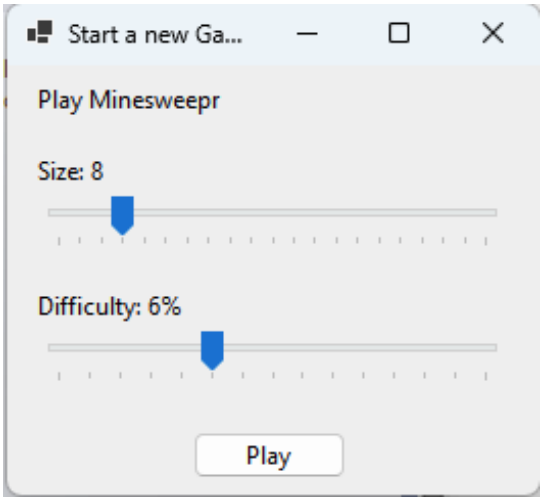

Figure 16. Form2 gets the size and difficulty values from the player.

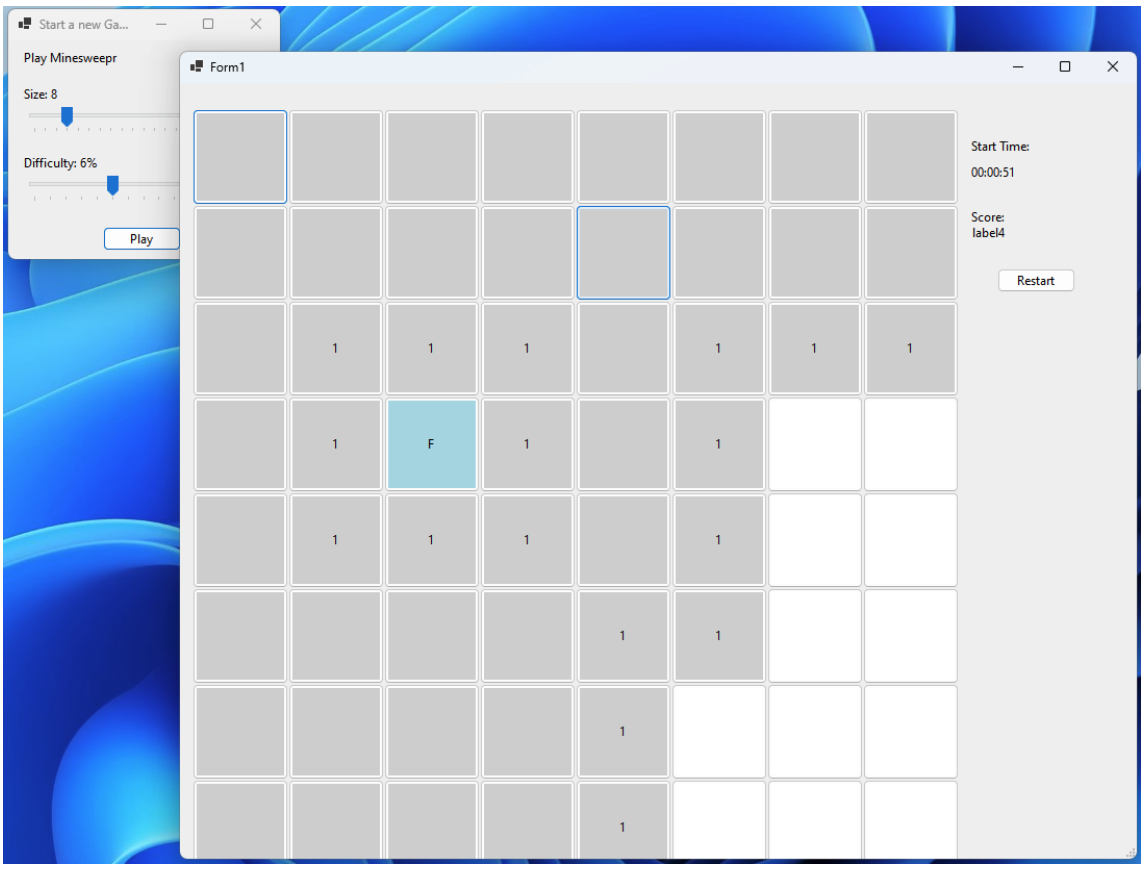

Figure 17. Game is in progress. Change button background colors and text properties to show the state of the cells.

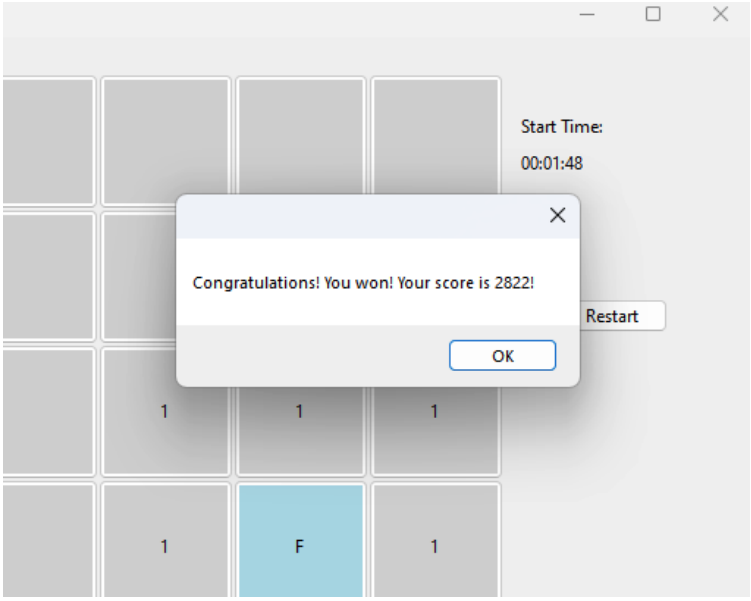

Figure 18. Game is over. Score is calculated based on size, difficulty, and time.

7. Add some game assets for Minesweeper graphics.

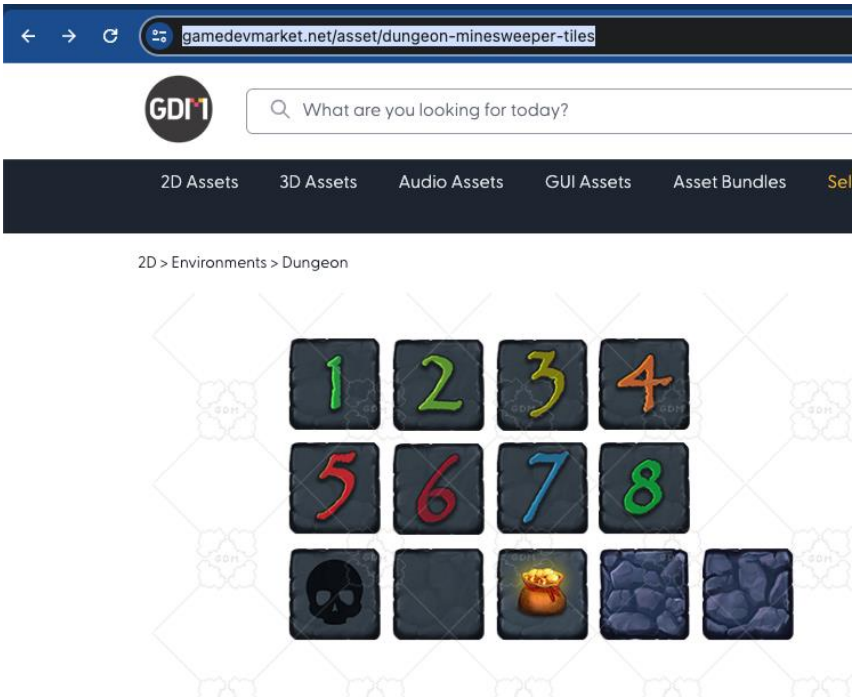

Figure 19. Free assets for a minesweeper game at GameDevMarket.com.

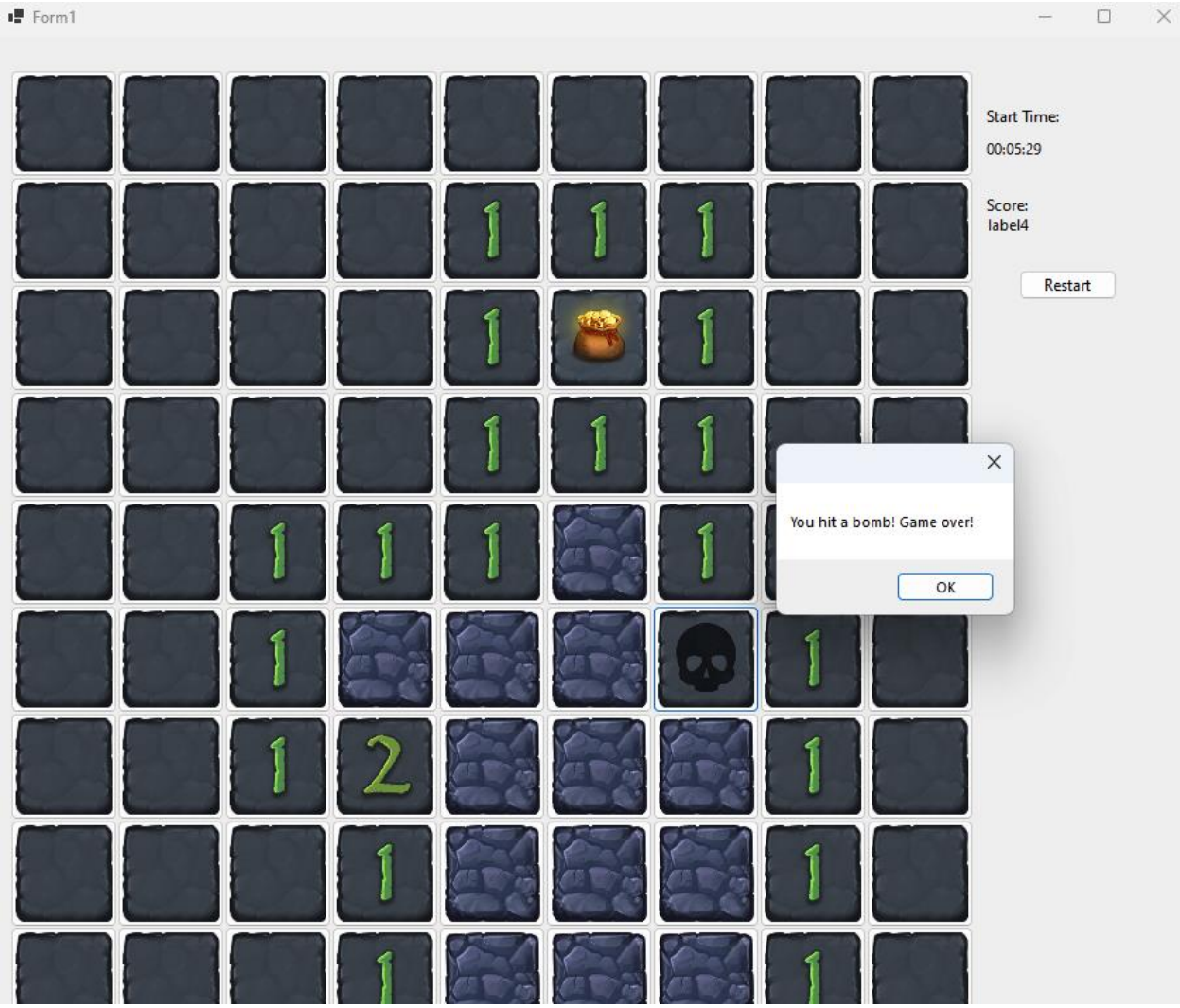

Figure 20. Game assets make the game look much better.

GitHub

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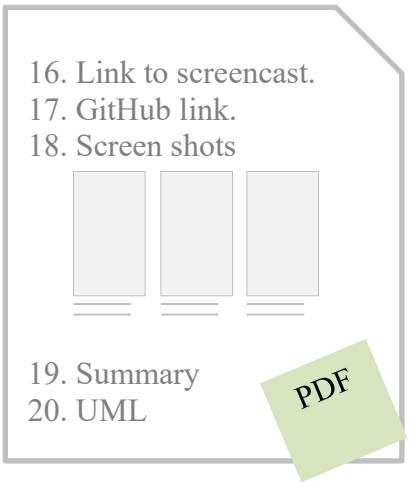
Deliverables



You must submit two separate files:

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- One ZIP file containing the complete codebase for Part 8.

Milestone Submission

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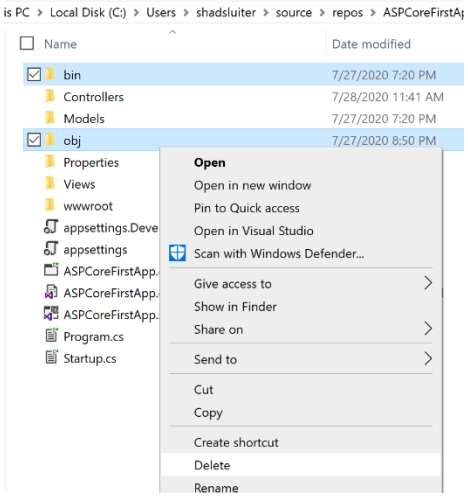

8. Submit a ZIP file of the project file. In order to save space, you can delete the bin and obj folders of the project. These folders contain the compiled version of the application and are automatically regenerated each time the build or run commands are executed.

Milestone 5: Implementing Game Statistics and High Score Tracking for Minesweeper

Overview

In this milestone, you will manage game stats for winning games.

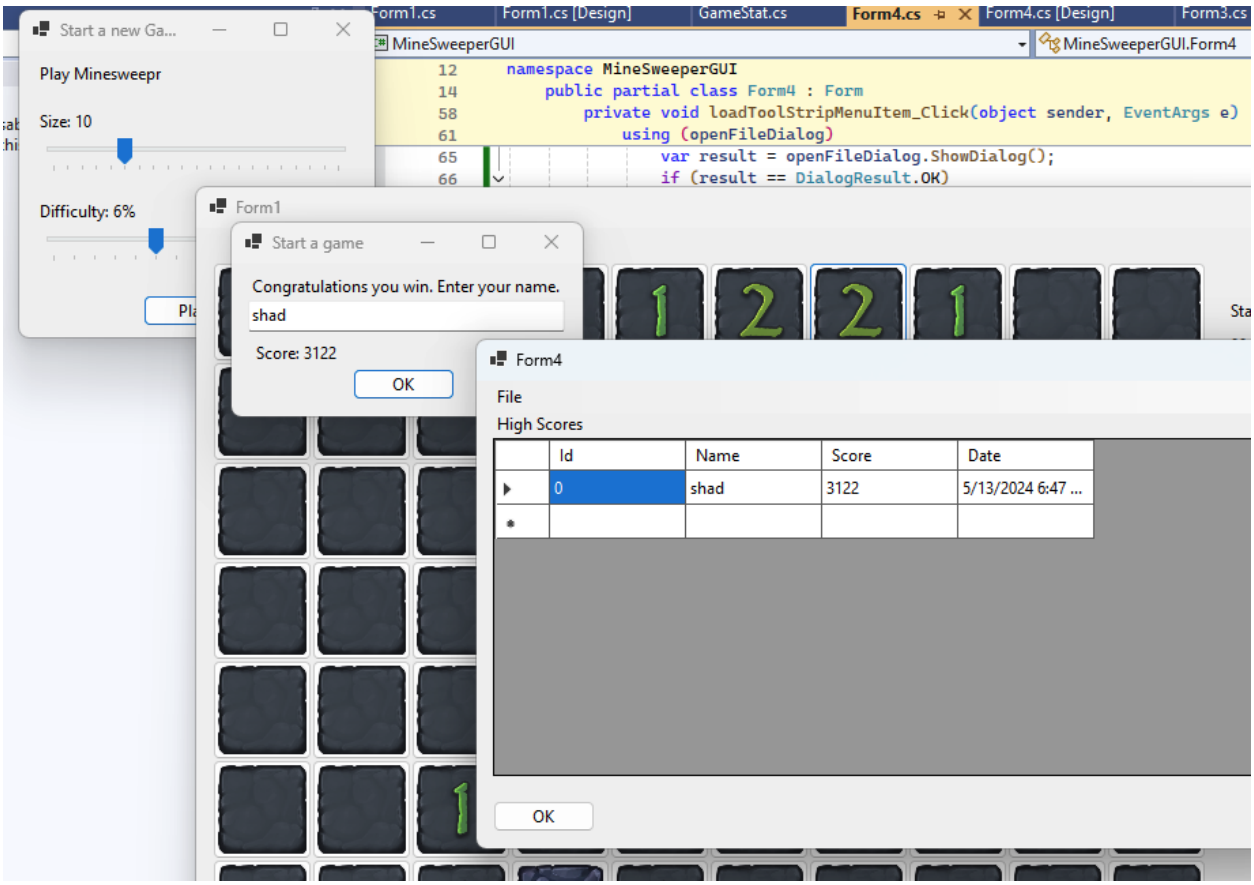

Figure 21. Form3 is used to capture the user's name. Form4 is used to show a list of scores in a grid view.

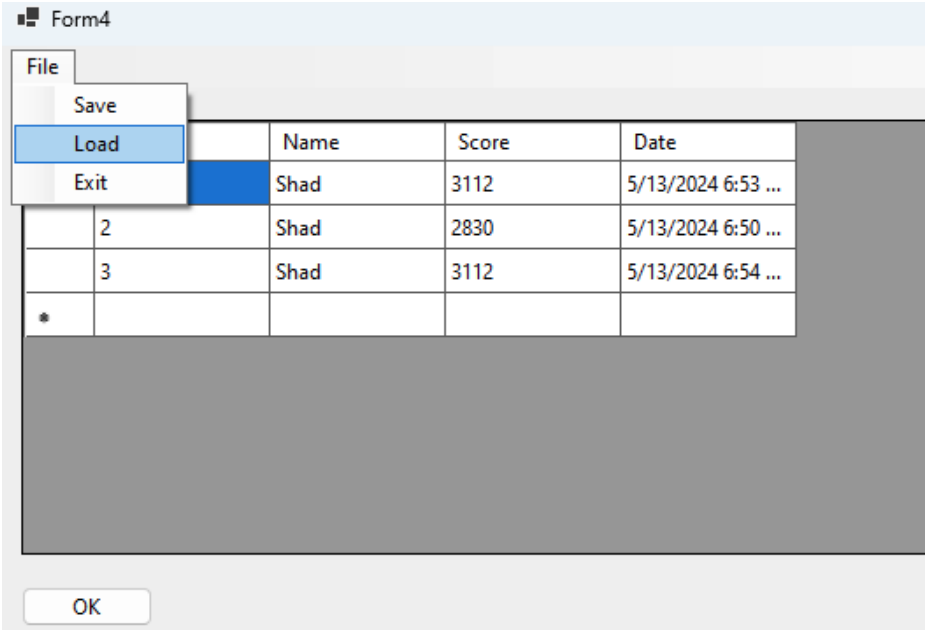

Figure 22. Form4 has "Save" and "Load" options to manage a text file that contains scores.

Execution

Execute this assignment according to the following guidelines:

1. Create a GameStat.cs class with properties id, name, score, and game time.
2. Create Form3 in order to capture the winner's name. Show Form3 after a winning game.

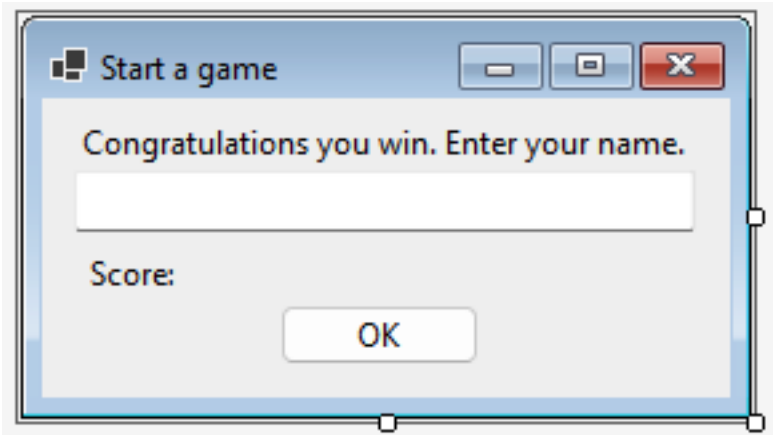

Figure 23. Get the user's name after winning.

3. Pass the **name** and **score** to Form4, which is used to display a list of winner's scores.

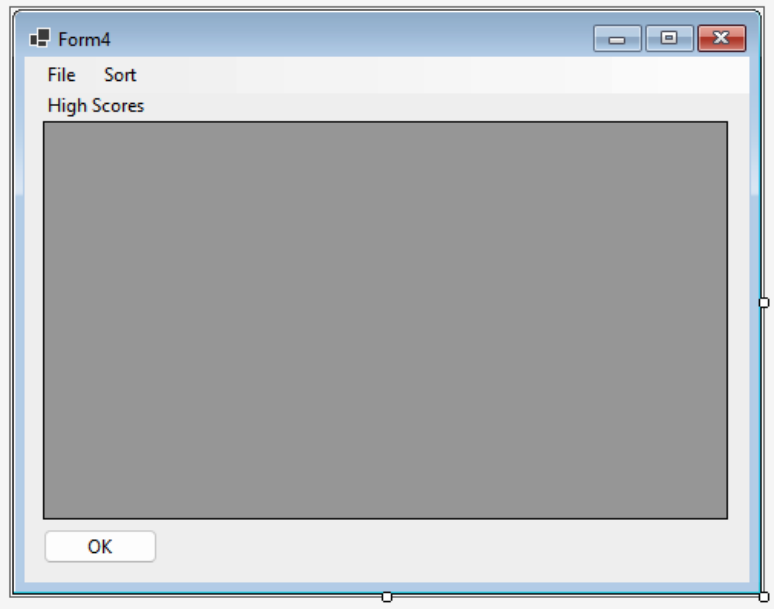

Figure 24. Form4 has a data grid view and two menu options.

4. Create a menu for Form4 that has a File > Save File > Load and File > Exit option.

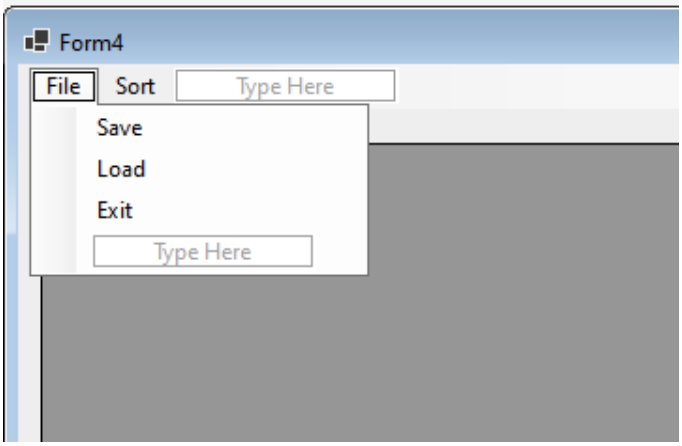

Figure 25. "File" menu has "Save," "Load," and "Exit" commands.

5. Add a menu for Sort > By Name, Sort > By Score, and Sort > By Date option.

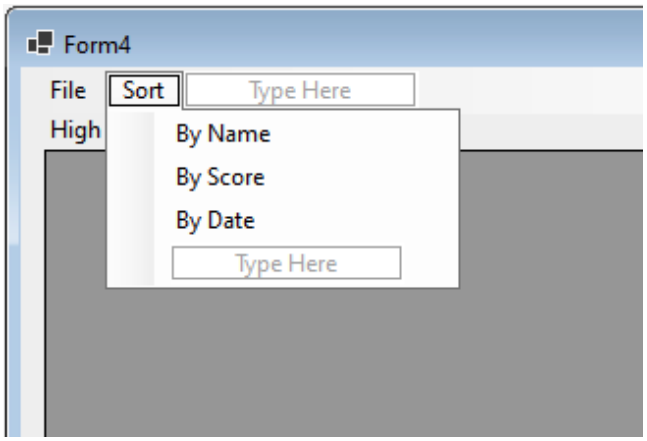

Figure 26. "Sort" menu has "By Name," "By Score," and "By Date" options.

6. Display the list of scores in the data grid control.

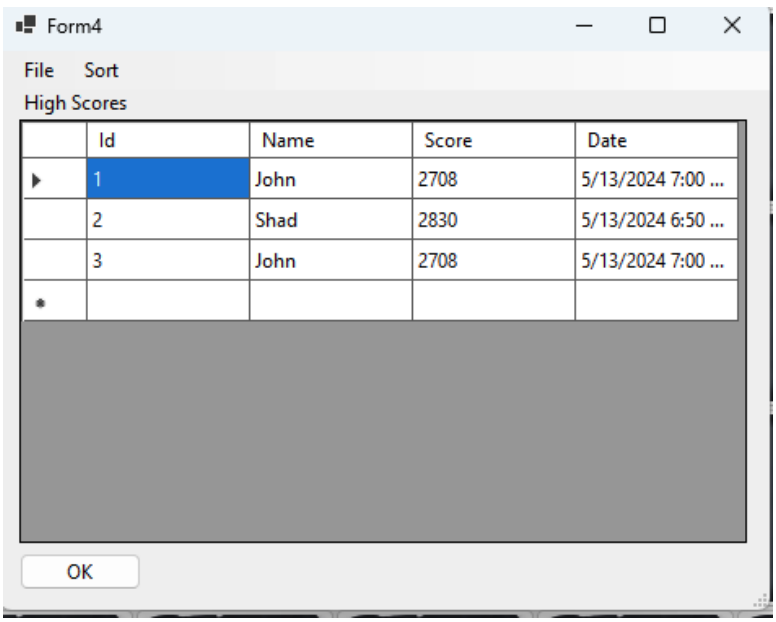

Figure 27. Three game stats have been played and saved to a text file.

7. Perform sorting operations according to the "Sort" menu choices.

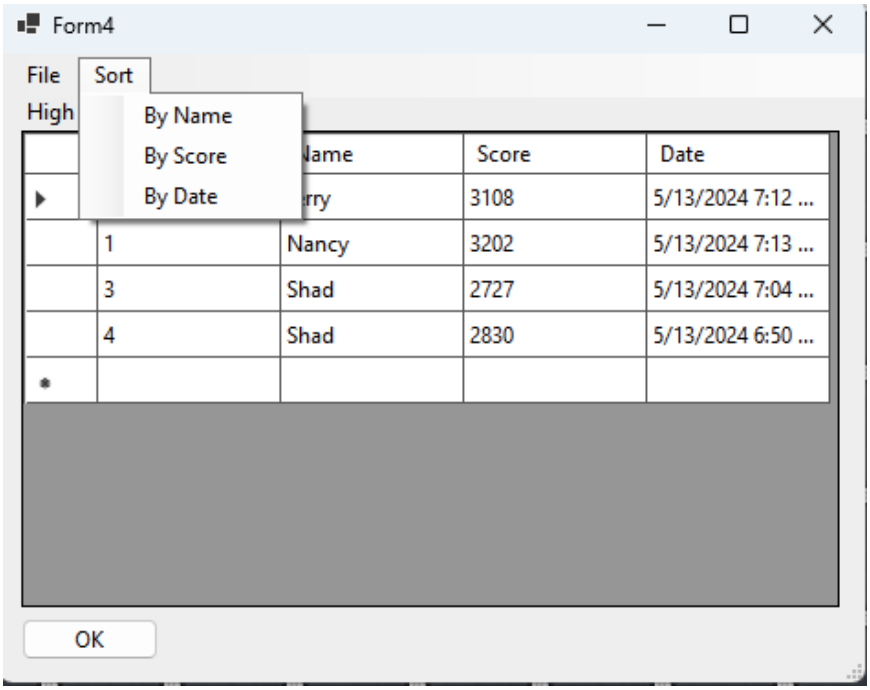

Figure 28. Sorting the scores by name, date, or score.

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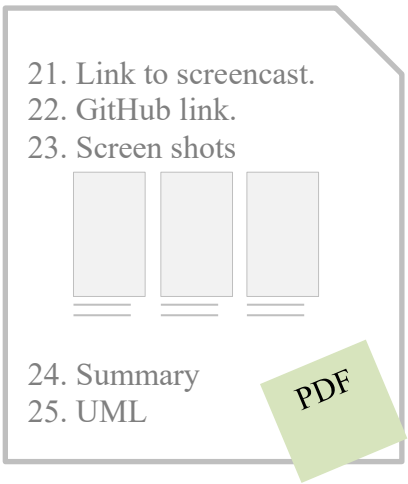

Milestone Submission

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c. This milestone must demonstrate the following pillars of object-oriented programming:

   i. Abstraction

   ii. Inheritance

   iii. Polymorphism

   iv. Encapsulation

d. Milestone must demonstrate the use of xUnit tests for BLL and must match UML diagrams.

e. Paste a link to the video in a Microsoft Word document at the top of the second page.

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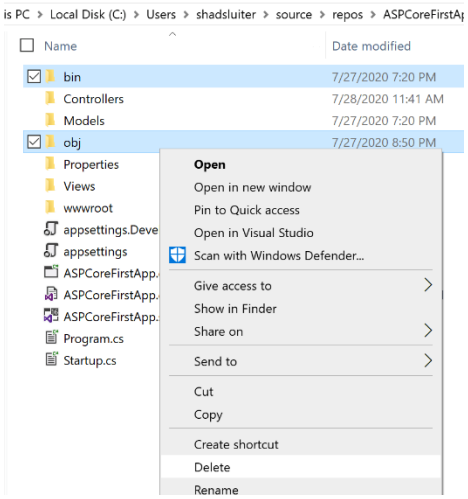

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Milestone 6: Extending Minesweeper with Advanced Features and Custom Enhancements

Objectives

1. Apply skills from previous lessons to an open-ended problem.
2. Design and test code to implement program requirements.

Choose two of the features listed below. Implement the suggestion. Document that the features work without errors.

1. **Special Reward Cells:** Review the special reward that has been part of the Cell class since Milestone 1. Implement the special power or reward that you defined.
2. **Custom Themes and Skins:** Allow players to customize the game's appearance by selecting different themes or skins for the UI, including the background, buttons, and icons. This could include seasonal themes or custom color schemes.
3. **Enhanced Stat Screen:** Enhance the PlayerStats form to include more analysis. Beneath the gridview, show average time per game, average number of points.
4. **Graphing:** Use a graphic library, like the one found here, to graphically display the range of scores on Form4. Show a line chart or bar chart to compare the scores earned in each game.
5. **Sound Effects and Music:** Integrate sound effects for game actions (flagging, revealing a cell, or hitting a bomb) and background music that can be toggled on or off. Provide a few music tracks to choose from, enhancing the gaming experience.
6. **Save and Resume Game:** Allow players to save their game progress and resume it later. Use the JSON serializing feature to save the Board class to a string and store in a text file. This feature would be especially useful for larger and more complex boards or when playing on higher difficulty levels.

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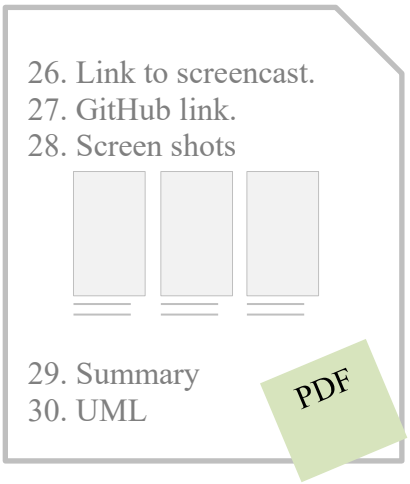

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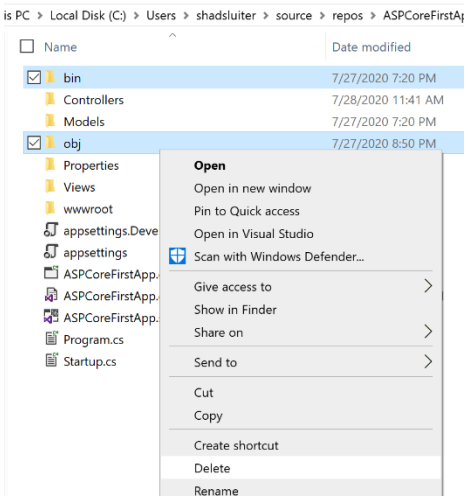

6. In the same document, in one paragraph, write a summary of the key concepts that were demonstrated in this lesson. Be sure to explain the key words introduced in this lesson.

7. In the same Word document, include the screenshots of the application being run successfully. Caption each image to explain what is being shown. Be sure to demonstrate the GUI version working successfully, then convert it to a PDF document.

8. Submit a ZIP file of the project file. In order to save space, you can delete the bin and obj folders of the project. These folders contain the compiled version of the application and are automatically regenerated each time the build or run commands are executed.